

Student Number	
Mark / 47	

Chemistry

**The Chemical Earth and Metals
Modules Test • 2003**

General Instructions

- Reading time – 5 minutes
- Working time – 55 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Student Number at the top of this page

Total Marks – 47

Part A – 12 marks

- Attempt Questions 1 – 12
- Allow about 10 minutes for this part

Part B – 35 marks

- Attempt Questions 13 – 23
- Allow about 45 minutes for this part

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Mark your answers for Questions 1 – 12 in the Answer Box on page 2.

- 1 Which of the following describes the atomic number of an element?
- (A) the number of protons plus neutrons in a neutral atom of the element
 - (B) the number of neutrons in a neutral atom of the element
 - (C) the number of protons in a neutral atom of the element
 - (D) the weighted mean of the isotopic masses of the element
- 2 Which of the following was the first alloy to be used by man?
- (A) brass
 - (B) bronze
 - (C) solder
 - (D) steel
- 3 Which of the following describes the periodic trend in the first ionisation energy of the elements?
- (A) decreases across a period
 - (B) increases as elements become more metallic
 - (C) decreases down a group
 - (D) increases with an increase in atomic number
- 4 Which of the following sets is composed exclusively of covalently bonded compounds?
- (A) Na_2O , N_2O_5 , NO
 - (B) Li_2O , Al_2O_3 , P_4O_{10}
 - (C) NO_2 , N_2O_5 , MgO
 - (D) NO_2 , SiO_2 , P_4O_{10}
- 5 The forces holding together atoms in solid xenon would be most similar to which of these?
- (A) the forces between the ions in NaCl
 - (B) the forces between the atoms in Cl_2
 - (C) the forces between the atoms in graphite
 - (D) the forces between the molecules of Br_2

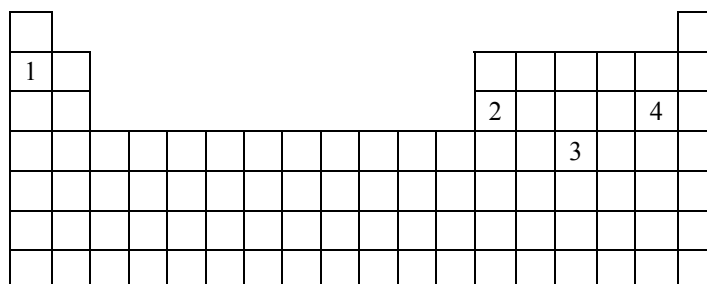
- 6

PHYSICAL PROPERTY	X	Y	Z
melting point (°C)	114	660	2370
electrical conductivity	nil	excellent	poor
ductile and malleable	nil	excellent	nil
metallic lustre	nil	yes	nil
density (g cm ⁻³)	4.90	2.69	2.30

Which classification of these elements given below is correct?

	X	Y	Z
(A)	non-metal	semi-metal	metal
(B)	non-metal	metal	semi-metal
(C)	semi-metal	metal	semi-metal
(D)	semi-metal	semi-metal	metal

- 7



Which numbers represent an element that is a semi-metal and an element that is a gas at 25°C and normal atmosphere pressure?

	Semi-metal	Gas at 25°C and normal atmosphere pressure
(A)	2	1
(B)	2	4
(C)	3	4
(D)	3	1

8 Which of the following is the best description of the structure and bonding in solid iodine at room temperature?

- (A) a network of iodine molecules with covalent bonding between the atoms, and weak intermolecular forces
- (B) a network of iodide ions held together by a 'sea' of delocalised electrons
- (C) a network of iodine atoms held together by covalent bonds in three dimensions
- (D) a network of iodide ions held together by ionic bonds in a 3-dimensional lattice

9 The diagram below shows a block of neighbouring elements on the periodic table.

K	Ca	Sc
Rb	Sr	Y
Cs	Ba	La

Which of the following elements would have an atomic radius greater than Sr?

- (A) La, Y
- (B) Cs, Rb
- (C) Ba, Y
- (D) Ca, Sc

10 The table shows the current cost of four metals which are mined and extracted in Australia.

METAL	PRICE
aluminium	\$2300 per tonne
gold	\$570 per 30 g
lead	\$750 per tonne
iron	\$1300 per tonne

Which of the following statements correctly identifies the cost factors of producing the metal?

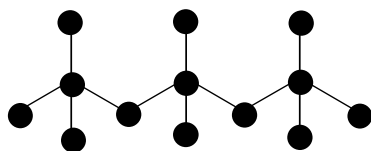
- (A) Aluminium ore is abundant but requires much energy to extract.
- (B) Gold ore is abundant but difficult to find and extract.
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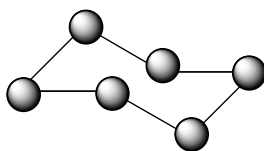
sphere	A	B	C	D
hydrosphere	seawater and fish	beach sand and gravel	seawater and fish	seawater and fish
biosphere	bacterial spores in air	seawater and fish	bacterial spores in air	seawater and fish
lithosphere	beach sand and gravel	beach sand and gravel	dusty air	dusty air
atmosphere	dusty air	bacterial spores in air	beach sand and gravel	bacterial spores in air

- (A) A
(B) B
(C) C
(D) D

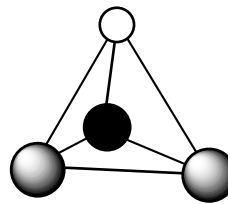
- 12 Study the following symbolic structures...



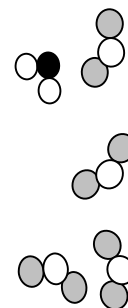
(i)



(ii)



(iii)



(iv)

Which of the following best describes the substances in i, ii, iii and iv?

Substance	A	B	C	D
i	compound	element	element	element
ii	compound	element	element	element
iii	compound	compound	element	compound
iv	compound	compound	compound	mixture

- (A) A
(B) B
(C) C
(D) D

Part B – 35 marks
Attempt Questions 13 – 23
Allow about 45 minutes for this part

Question 13 (6 marks)

Write balanced formulae equations for the following reactions.

(a) calcium + water

(b) magnesium + dilute hydrochloric acid

(c) sodium + oxygen

Write a pair of half-equations for the following reactions.

(d) aluminium + dilute nitric acid

(e) potassium + dilute sulfuric acid

(f) barium + dilute hydrochloric acid

Question 14 (7 marks)

- (a) Using the information in the table, classify the substances (A, B, C, D) as either metallic, ionic, covalent network or covalent molecular structures. **(2 marks)**

	SUBSTANCE			
PROPERTY	A	B	C	D
melting point (°C)	3350	– 114	961	801
soluble in water	no	yes	no	yes
solid conducts electricity	no	no	yes	no
molten (liquid) state conducts electricity	no	no	yes	yes
hardness	very hard	solid is soft	hard	hard
other properties	crystalline solid	aqueous solution conducts	shiny solid	–

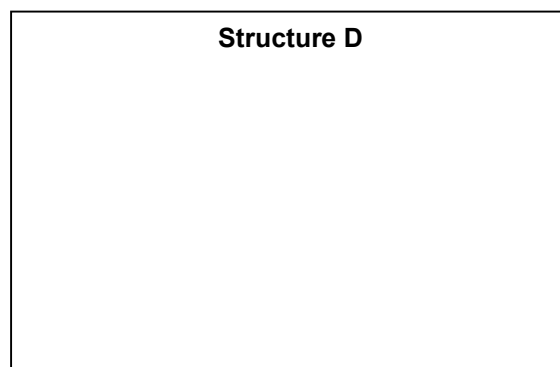
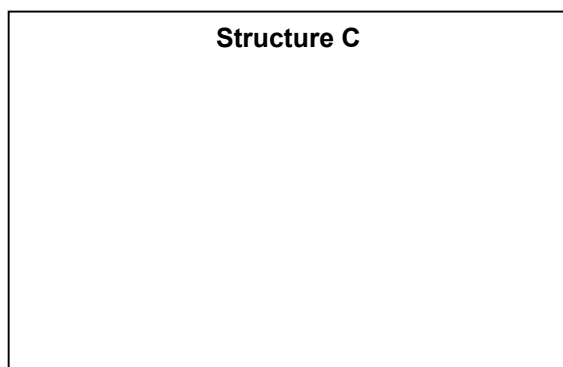
Substance A _____

Substance B _____

Substance C _____

Substance D _____

- (b) Distinguish between structures C and D by drawing a diagram showing the arrangement and type of particles in each structure. **(2 marks)**



Question 14 continues on page 9

Question 14 (continued)

- (c) Explain the difference in electrical conductivity between substances C and D. **(2 marks)**

- (d) State the type of particles found in substance A and the type of bonding between these particles. **(1 mark)**

Question 15 (3 marks)

Name the compounds listed below.

(a) NaHCO_3 _____

(b) SF_6 _____

(c) Ba(OH)_2 _____

Give the formulae for the compounds listed below.

(d) ammonium nitrate _____

(e) dinitrogen pentoxide _____

(f) aluminium sulfate _____

Question 16 (3 marks)

(a) Draw the Lewis electron dot structure for the oxide ion. **(1 mark)**

(b) Give the electronic configuration for the oxide ion. **(1 mark)**

(c) Write the name of another species with the same electronic configuration as the oxide ion. **(1 mark)**

Question 17 (2 marks)

(a) Write a balanced formulae equation for the thermal decomposition of copper(II) carbonate. **(1 mark)**

(b) Describe a chemical test that can be used to prove that copper(II) carbonate can be decomposed by heat.

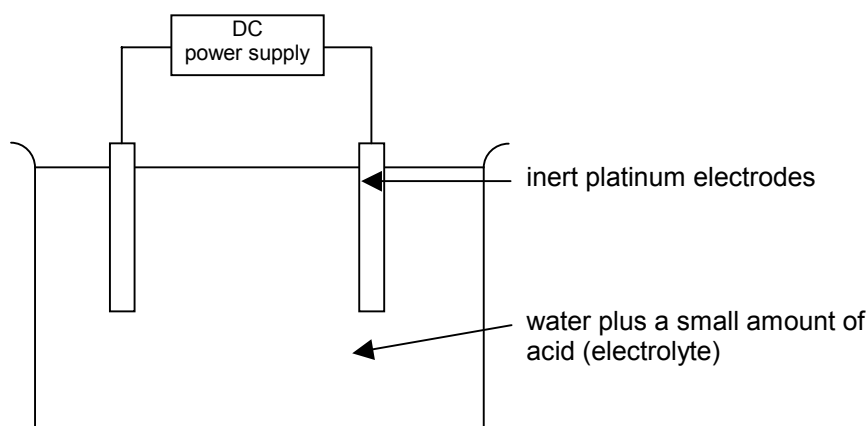
(1 mark)

Question 18 (1 mark)

Identify an example that demonstrates that light can be absorbed during the decomposition of a compound.

Question 19 (2 marks)

The diagram shows the set-up for a laboratory apparatus in which two inert (non-reactive) platinum electrodes are placed in a beaker of water that has a small amount of acid added to it.



Describe what would be observed at the electrodes when electricity flows into the the electrodes.
Give a reason why this happens.

Question 20 (1 mark)

Why is the amount of energy needed to break the chemical bond in a hydrogen molecule greater than the energy needed to break the chemical bond in a hydrogen iodide molecule?

Question 21 (4 marks)

An analytical chemist used fractional distillation to analyse a crude oil mixture from a newly discovered oil well. A 60 g sample of the crude oil was weighed accurately and then fractionally distilled.

Five fractions: A, B, C, D and E were isolated and weighed. The weights obtained in grams were: 2, 18, 9, 12, 16 for components A, B, C, D and E respectively.

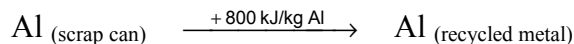
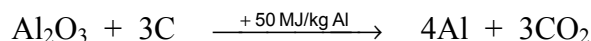
(a) Construct a table of results for the data given. **(1 mark)**

(b) To be an economically productive source, component B must be at least 35% by mass of the crude oil. Assess the potential economic productivity of this sample of crude oil. Justify your answer by including working. **(2 marks)**

(c) Describe **another** situation in which gravimetric analysis supplies useful data for scientists. **(1 mark)**

Question 22 (1 mark)

The recycling of aluminium cans is a great success in Australia with over 50% of cans being collected and recycled! Study the equations showing the extraction and recycling of aluminium...



Account for the huge difference in the energy requirement based on the nature of the processes involved.

Question 23 (5 marks)

(a) Identify the original data which Mendeleev used to formulate the Periodic Table.

(i) Physical data **(1 mark)** _____

(ii) Chemical data **(1 mark)** _____

(b) Historically, Mendeleev made accurate predictions about element 32 (germanium) before it was discovered. According to the HSC Periodic Table, element 117 has not yet been synthesised; however, we also can make predictions about its properties in relation to its neighbouring elements based on periodic law principles.

(i) Compare the relative electronegativity of element 117 with that of ununoctium (element 118).

(1 mark) _____

(ii) Compare the valency of element 117 with that of astatine.

(1 mark) _____

(iii) Compare the relative reactivity of element 117 with that of astatine.

(1 mark) _____

DATA SHEET

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 101.3 kPa (1.00 atm) and	
at 273 K (0°C)	22.41 L
at 298 K (25°C)	24.47 L
Ionisation constant for water at 298 K (25°C), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (4th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

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1 H 1.008 Hydrogen																		He 4.003 Helium			
		KEY																			
		Atomic Number																			
		Symbol of element																			
		Atomic Weight																			
		Name of element																			

KEY

Atomic Number	Symbol of element
79	Au
Atomic Weight	Name of element
197.0	Gold

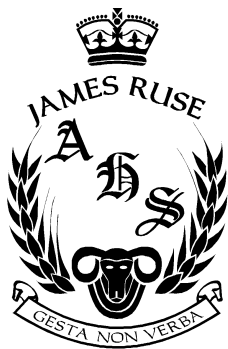
Lanthanides

57	La	58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb	66	Dy	67	Ho	68	Er	69	Tm	70	Yb	71	Lu
138.9		140.1		140.9		144.2		[146.9]		150.4		152.0		157.3		158.9		162.5		164.9		167.3		168.9		173.0		175.0	
Lanthanum		Cerium		Praseodymium		Neodymium		Promethium		Samarium		Europium		Gadolinium		Terbium		Dysprosium		Holmium		Erbium		Thulium		Ytterbium		Lutetium	

Actinides

89	Ac	90	Th	91	Pa	92	U	93	Np	94	Pu	95	Am	96	Cm	97	Bk	98	Cf	99	Es	100	Fm	101	Md	102	No	103	Lr
[227.0]		232.0		231.0		238.0		[237.0]		[239.1]		[241.1]		[244.1]		[249.1]		[252.1]		[252.1]		[257.1]		[258.1]		[259.1]		[262.1]	
Actinium		Thorium		Protactinium		Uranium		Neptunium		Plutonium		Americium		Curium		Berkelium		Californium		Einsteinium		Fermium		Mendelevium		Nobelium		Lawrencium	

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.



Marking Scheme and Outcomes

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correct (with an arrow pointing to B)

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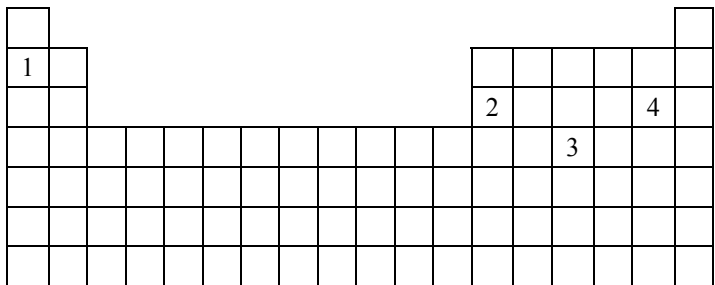
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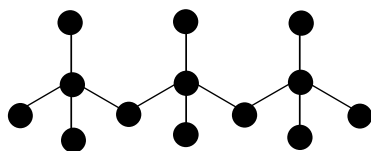
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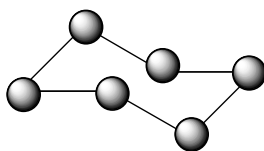
sphere	A	B	C	D
hydrosphere	seawater and fish	beach sand and gravel	seawater and fish	seawater and fish
biosphere	bacterial spores in air	seawater and fish	bacterial spores in air	seawater and fish
lithosphere	beach sand and gravel	beach sand and gravel	dusty air	dusty air
atmosphere	dusty air	bacterial spores in air	beach sand and gravel	bacterial spores in air

- (A) A
(B) B
(C) C
(D) D

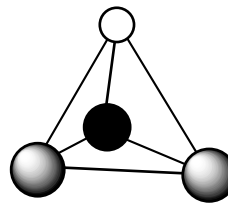
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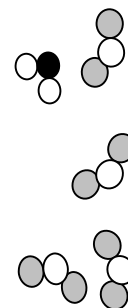
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ii	compound	element	element	element
iii	compound	compound	element	compound
iv	compound	compound	compound	mixture

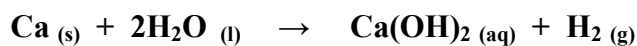
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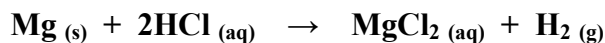
Question 13 (6 marks)

Write balanced formulae equations for the following reactions. **OUTCOME – P 13**
(1 mark each; lack of subscripts – minus 1 mark max.)

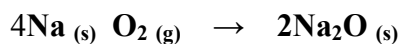
(a) calcium + water



(b) magnesium + dilute hydrochloric acid

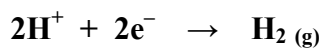


(c) sodium + oxygen

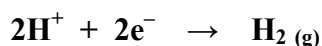
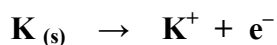


Write a pair of half-equations for the following reactions. **OUTCOME – P 13**
(1 mark per pair; maximum of 1 mark lost for reduction half-equation)

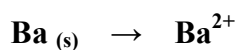
(d) aluminium + dilute nitric acid



(e) potassium + dilute sulfuric acid



(f) barium + dilute hydrochloric acid



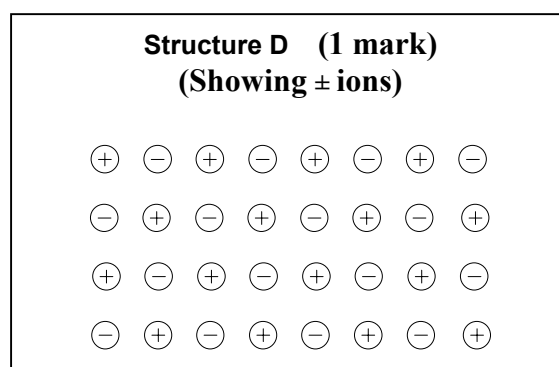
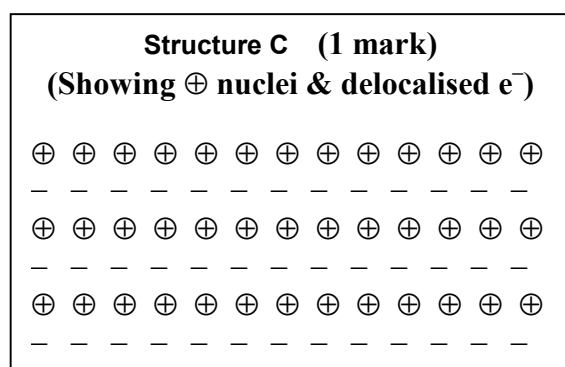
Question 14 (7 marks)

- (a) Using the information in the table, classify the substances (A, B, C, D) as either metallic, ionic, covalent network or covalent molecular structures. **OUTCOME – P 14 (2 marks)**

	SUBSTANCE			
PROPERTY	A	B	C	D
melting point (°C)	3350	– 114	961	801
soluble in water	no	yes	no	yes
solid conducts electricity	no	no	yes	no
molten (liquid) state conducts electricity	no	no	yes	yes
hardness	very hard	solid is soft	hard	hard
other properties	crystalline solid	aqueous solution conducts	shiny solid	–

Substance A **covalent network**
 Substance B **covalent molecular**
 Substance C **metallic lattice**
 Substance D **ionic lattice**

- (b) Distinguish between structures C and D by drawing a diagram showing the arrangement and type of particles in each structure. **OUTCOME – P 13 (2 marks)**



Question 14 continues on page 9

Question 14 (continued)

- (c) Explain the difference in electrical conductivity between substances C and D. **(2 marks)**

OUTCOME – P 14

Substance C conducts electricity in solid and molten states due to free (delocalised) electrons which can move freely through the lattice.

Substance D does not conduct electricity in the solid state because ions are fixed. However, when molten, the ions are mobile, free to move and conduct electricity.

- (d) State the type of particles found in substance A and the type of bonding between these particles.

OUTCOME – P 14 (1 mark)

Atoms; covalent bonding. (N.B. Must state both)

Question 15 OUTCOME – P 6 (3 marks)

Name the compounds listed below. **(3 @ ½ mark)**

- (a) NaHCO_3 **sodium hydrogen carbonate or sodium bicarbonate**

- (b) SF_6 **sulfur hexafluoride**

- (c) Ba(OH)_2 **barium hydroxide**

Give the formulae for the compounds listed below. **(3 @ ½ mark)**

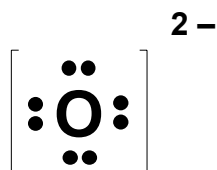
- (d) ammonium nitrate **NH_4NO_3**

- (e) dinitrogen pentoxide **N_2O_5**

- (f) aluminium sulfate **$\text{Al}_2(\text{SO}_4)_3$**

Question 16 OUTCOME – P 6 (3 marks)

- (a) Draw the Lewis electron dot structure for the oxide ion. **(1 mark)**



- (b) Give the electronic configuration for the oxide ion. **(1 mark)**

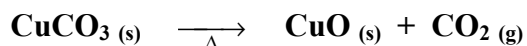
2 – 8 or 2,8

- (c) Write the name of another species with the same electronic configuration as the oxide ion. **(1 mark)**

neon, fluoride ion, sodium ion, etc.

Question 17 OUTCOME – P 8 (2 marks)

- (a) Write a balanced formulae equation for the thermal decomposition of copper(II) carbonate. **(1 mark)**



- (b) Describe a chemical test that can be used to prove that copper(II) carbonate can be decomposed by heat. **(1 mark)**

CO₂ is a product of the decomposition of CuCO₃ and this can react with limewater forming a white precipitate.

– OR –

CuO is a product which reacts with dilute H₂SO₄ forming a blue solution (CuSO₄).

Question 18 OUTCOME – P 7 (1 mark)

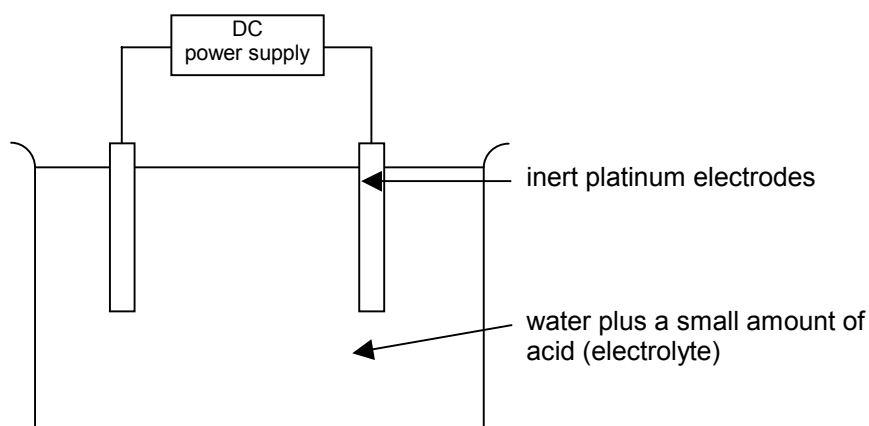
Identify an example that demonstrates that light can be absorbed during the decomposition of a compound.

Silver salts (AgNO₃, AgCl, AgBr, AgI) can be decomposed using light energy.

e.g. AgNO₃ forms a black colour as a result of the Ag⁺ forming Ag_(s) and thus decomposing the AgNO₃.

Question 19 OUTCOME – P 7 (2 marks)

The diagram shows the set-up for a laboratory apparatus in which two inert (non-reactive) platinum electrodes are placed in a beaker of water that has a small amount of acid added to it.



Describe what would be observed at the electrodes when electricity flows into the the electrodes.
Give a reason why this happens.

Gas bubbles will be observed at both electrodes. (1 mark)

The gases are hydrogen and oxygen and these are a result of the electrolysis of water. (1 mark)

Question 20 (1 mark)

Why is the amount of energy needed to break the chemical bond in a hydrogen molecule greater than the energy needed to break the chemical bond in a hydrogen iodide molecule?

The H – H bond is stronger than the H – I bond.

Question 21 OUTCOMES – P 2, 10, 14 (4 marks)

An analytical chemist used fractional distillation to analyse a crude oil mixture from a newly discovered oil well. A 60 g sample of the crude oil was weighed accurately and then fractionally distilled. Five fractions: A, B, C, D and E were isolated and weighed. The weights obtained in grams were: 2, 18, 9, 12, 16 for components A, B, C, D and E respectively.

- (a) Construct a table of results for the data given. **(1 mark for whole table)**

Component	Weight (g)
A	2
B	18
C	9
D	12
E	16

- (b) To be an economically productive source, component B must be at least 35% by mass of the crude oil. Assess the potential economic productivity of this sample of crude oil. Justify your answer by including working. **(2 marks)**

According to the criterion set, the oil well will not be economically productive. (1 mark)

$$\text{Percentage of B} = \frac{\text{mass B}}{\text{mass sample}} \times 100\%$$

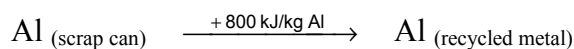
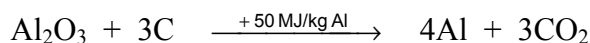
(1 mark)

$$\text{Percentage of B} = \frac{18 \text{ g}}{60 \text{ g}} \times 100\% = 30\% \text{ by mass}$$

- (c) Describe **another** situation in which gravimetric analysis supplies useful data for scientists. **(1 mark)**
- To determine the composition of soil in a particular location to see if it is suitable for growing certain crops.
 - To determine the amounts of particular substances present in water or air to decide how polluted the samples are.
 - To decide whether a particular commercial mixture being sold has the same percentage composition as a similar mixture marketed by a rival company.
 - Other possible situations.

Question 22 OUTCOME – P 4 (1 mark)

The recycling of aluminium cans is a great success in Australia with over 50% of cans being collected and recycled! Study the equations showing the extraction and recycling of aluminium...



Account for the huge difference in the energy requirement based on the nature of the processes involved.

The extraction of aluminium from Al_2O_3 involves a chemical reaction... the breaking of the strong bond with oxygen.

The recycling of aluminium involves melting which is a physical change requiring much less energy.
– OR –

The recycling of aluminium involves the breaking of the metallic bonding in the scrap Al.

This bond is weaker than the bond existent within Al_2O_3 , $\therefore$ less energy is required.

Question 23 OUTCOME – P 6 (5 marks)

- (a) Identify the original data which Mendeleev used to formulate the Periodic Table.
- (i) Physical data (1 mark) **Atomic weight, density, atomic volume (any one)**
 - (ii) Chemical data (1 mark) **Formulae of elemental oxides, chlorides, hydrides; valency; reaction with acid/alkali (any one)**
- (b) Historically, Mendeleev made accurate predictions about element 32 (germanium) before it was discovered. According to the HSC Periodic Table, element 117 has not yet been synthesised; however, we also can make predictions about its properties in relation to its neighbouring elements based on periodic law principles.
- (i) Compare the relative electronegativity of element 117 with that of ununoctium (element 118).
(1 mark) 117 > 118
 - (ii) Compare the valency of element 117 with that of astatine.
(1 mark) 117 = astatine
 - (iii) Compare the relative reactivity of element 117 with that of astatine.
(1 mark) 117 < astatine

DATA SHEET

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 101.3 kPa (1.00 atm) and	
at 273 K (0°C)	22.41 L
at 298 K (25°C)	24.47 L
Ionisation constant for water at 298 K (25°C), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (4th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

1		KEY																2																	
H	1.008																	He	4.003																
Hydrogen																		Helium																	
3		4																		10															
Li	6.941	Be	9.012																	Ne	20.18														
Lithium		Beryllium																		Neon															
11		12																		18															
Na	22.99	Mg	24.31																	Ar	39.95														
Sodium		Magnesium																		Argon															
19		20		21		22		23		24		25		26		27		28		29		30		31		32		33		34		35		36	
K	39.10	Ca	40.08	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr																
Potassium	Calcium	Scandium	Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel	Copper	Zinc	Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton																		
37		38		39		40		41		42		43		44		45		46		47		48		49		50		51		52		53		54	
Rb	85.47	Sr	87.62	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe																
Rubidium	Strontium	Yttrium	Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium	Silver	Cadmium	Indium	Tin	Antimony	Tellurium	Iodine	Xenon																		
55		56		57-71		72		73		74		75		76		77		78		79		80		81		82		83		84		85		86	
Cs	132.9	Ba	137.3	Lanthanides		Hf	Ta	W	Re	Os	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn																
Cesium	Barium	Lanthanides		Hafnium	Tantalum	Tungsten	Rhenium	Osmium	Iridium	Platinum	Gold	Mercury	Thallium	Lead	Bismuth	Polonium	Astatine	Radon																	
87		88		89-103		104		105		106		107		108		109		110		111		112		113		114		115		116		117		118	
Fr	[223.0]	Ra	[226.0]	Actinides		Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub	Uuq	Uuh	Uus	Uuo																	
Francium	Radium	Actinides		Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Ununilium	Unununium	Ununbium	Ununtrium	Ununquadium	Ununpentium	Ununhexium	Ununseptium	Ununoctium																	

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
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Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.